

★ Percentage

Per Cent
(Hundred)

$$50\% + 50\% \Rightarrow 75\%$$

$$50\% + 20\% \Rightarrow 60\%$$

of ① $50\% + 50\% = 75\%$

$$\downarrow \quad \downarrow$$

$$0.5 \times 0.5 = 0.25$$

If the value less than 1, decreasing
If the value more than 1, increasing

② $50\% \downarrow + 20\% \downarrow = 60\% \downarrow$

$$(100-50) \text{ into decimal } \downarrow \quad \downarrow$$

$$0.5 \times 0.8 = 0.4$$

③ $50\% \uparrow + 50\% \downarrow = 25\% \downarrow$

$$(100+50) \text{ into decimal } \downarrow \quad \downarrow$$

$$1.5 \times 0.5 = 0.75$$

→ (less than 1)

→ How much less than 1
 $\Rightarrow 0.25$
 $\Rightarrow 25\% \downarrow$

④ $10\% \uparrow + 10\% \uparrow + 10\% \uparrow$

$$1.1 \times 1.1 \times 1.1 = 1.331 \uparrow$$

→ How much more than 1
 $\Rightarrow 33\%$
 $\Rightarrow 33\% \uparrow$

Pattern 1:

⇒ Successive

Increase/Decrease

Que

when the price of mobile reduced by 20%. the number of mobile sold increase by 40%. The effect on the revenue was?

- (A) 12% increase (B) 12% decrease (C) 32% increase
(D) 40% decrease

$$20\% \downarrow + 40\% \uparrow =$$

$$0.8 \times 1.4 = 1.12 \uparrow \text{ (more than 1)}$$

12% ↑ increase

Que A trader offers three successive discounts of 20%, 10% and 5% to customer. How much is overall single discount.
 (A) 30% (B) 31.6% (C) 35% (D) 68.4%
Sol $20\% \downarrow + 10\% \downarrow + 5\% \downarrow$
 $0.8 \times 0.9 \times 0.95 = 0.684 \times 100 \Rightarrow 68.4\%$
 Discount = $100 - 68.4\%$
 $= 31.6\%$

Que 8% of the people eligible to vote are between 20 & 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25, ~~also~~ actually voted. In that election number of person between 20 and 25, who actually voted, was what percentage of those eligible to vote?
 (A) 4.2% (B) 8% (C) 6.4% (D) 6.8%

Percentage :=
 $\Rightarrow$ Successive INC/DEC
 $\Rightarrow$ Comparison (%)

Sol $0.85 \times 0.08 \times \text{Total eligible votes}$
 $\Rightarrow 0.068 \text{ of total eligible votes}$
 $\times 100$
 $= 6.8\%$
 Note: If we are asked to decrease the value then we subtract from 1, else not.

Pattern 2 [Comparison]

Que If M is 25% more than N, then N is how much percent less than M?

Sol $\nwarrow$ 25% more
 $M = 125\% N$
 $N = \frac{M}{125\%}$
 $N = M \times \frac{100}{125}$
 $[N = M \times 0.8]$
 Less than 1
 $\Rightarrow$ How much less $\Rightarrow 100 - 0.8$
 $\Rightarrow 0.2$
 ($\therefore$ 20% less) ANS.

Que If the petrol rate is increased by 40%, then how much percentage we should decrease our consumption, in order to maintain same budget.

(A) 71.42% (B) 40% (C) 28.57% (D) 60%

Sol

$$\text{Increase} \Rightarrow 40\% \Rightarrow \frac{140}{100}$$

Moving this side

$$= \frac{140}{100}$$

$\Rightarrow 71.43\%$ Consumption.

$$\frac{5}{7} \frac{100}{140}$$

$$= 0.7143$$

$\Rightarrow$ less than 1

How much less than 1

$$\Rightarrow 1 - 0.7143 = 0.2857$$

$$\Rightarrow 28.57\%$$

Que If the length of rectangle increase by 20%, then how much percent we should decrease the breadth in order to maintain same area.

Sol (A) 20% (B) 16.67% (C) 83.33% (D) 81.33%

$$\text{Inc} \Rightarrow 20\% \Rightarrow \frac{120}{100}$$

$$\frac{5}{6} \frac{100}{120}$$

$$\Rightarrow 0.8333$$

$$\Rightarrow 83.33\%$$

$\Rightarrow$ less than 1

How much less than 1

$$\Rightarrow 1 - 0.8333 \Rightarrow 0.1666$$

$$\Rightarrow 16.67\%$$

Assignment

Que If the side of square is increased by 20%, then what is the percentage change in its area?

(A) 44% (B) 80% (C) 22% (D) 144%

Sol

$$\text{Inc} \Rightarrow 20\% \Rightarrow \frac{120}{100}$$

$$\therefore \text{Area} = 1.2 \times 1.2$$

$$= 1.44$$

$$\Rightarrow 1.44 \text{ Increase inside}$$

$$\Rightarrow 1.44$$

$$\Rightarrow 44\%$$

Que The population of a town doubled every 5 years from 2000 to 2015. What is the percentage increase in population in this period?
 (A) 800%. (B) 400%. (C) 700%. (D) 600%.

Sol

100% $\rightarrow$ 200% (2000-2005)
 200 $\rightarrow$ 400% (2005-2010)
 400 $\rightarrow$ 800% (2010-2015)

100% at first
 800% current
 $\downarrow$
 700% increase

Que If P is 60% taller than Q, by what percent is Q shorter than P?
 (A) 40%. (B) 37.5%. (C) 62.5%. (D) None of these.

Sol

P $\rightarrow$ 60% $\uparrow$ $\Rightarrow$ 160% Q

$\frac{P}{160\%} \Rightarrow Q$

$\frac{5}{8} \times \frac{100}{100} \times P \Rightarrow Q$

$P \times 0.625 \Rightarrow Q$

less than 1
 How much

$1 - 0.625$

$\Rightarrow 0.375$

$\Rightarrow 37.5\%$

Que 5% of income of P is equal to 15% of income of Q and 10% of income of Q equal to 20% of income of R. If R's income is 2000, then what is total income of P, Q and R?
 (A) 9000 (B) 12000 (C) 15000 (D) 18000

Sol

R's income $\Rightarrow$ 2000

(R) 20% $\Rightarrow$ 400

(Q) 10% $\Rightarrow$ 400

(Q) 100% $\Rightarrow$ 4000

15% of 4000

4000 $\rightarrow$ 100%

(1) $\leftarrow$ 15%

600 (15%) Q

$P(5\%) = Q(15\%)$

$P(5\%) = 600$

$\Rightarrow 5\% = 600$

100% $\Rightarrow$ (1)

(P) 12000

R = 2000
 Q = 4000
 P = 12000

18000

Percentage

$$\text{Error \%} = \frac{\text{diff}}{\text{Actual Value}} \times 100$$

$$\Rightarrow \frac{9}{45} \times 100$$

Data enter $\Rightarrow 45 \leftarrow$ Actual value.

$\rightarrow$ Type 54 $\leftarrow$ Wrong value.

$$54 - 45 \Rightarrow 9 \text{ (diff)}$$

$$\text{Correction \%} = \frac{\text{diff.}}{\text{Wrong Value}} \times 100$$

Ques Sanjeev saves 20% of his income. If his income is increased by 20% and expenditure decrease by 10%, then find the percentage change in his savings.

Income Saving Expenditure

100 20 80

120 48 72

increase. Actual saving = 20

$$\text{Diff} \Rightarrow 48 - 20 = 28$$

$$\frac{28}{20} \times 100 = 140\% \uparrow$$

Profit & Loss
(Based on Percentage)

C.P

S.P

M.P